Stereo-thermal-lithography: a new principle for rapid prototyping

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Keywords

Stereolithography, Polymers, Fabrication

Abstract

A new fabrication process for rapid prototyping is proposed in this paper. Optical and thermal effects are simultaneously used in this process to locally induce a phase change in a liquid resin. This phase change phenomena is used to "write" three-dimensional shapes or patterns. Such objects or patterns can involve macroscopic engineering prototypes through to nanostructures for exploitation in waveguiding and photonic crystals. Several advantages can be achieved through this new process, in terms of accuracy, cost and time.

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Introduction

The most important rapid prototyping technologies currently available or under research are shown in Figure 1. Among them, stereolithography (SL) (Hull, 1986) is the most popular process. SL involves the curing or solidification of a liquid photosensitive polymer by a laser beam scanned across its surface. The laser supplies energy that induces a chemical reaction, bonding large number of small molecules and forming a highly cross-linked polymer. As the reaction proceeds, the viscosity of the resin in the direct vicinity of the light spot increases until vitrification takes place, freezing the curing process and promoting an incomplete curing reaction. The residual unreacted polymer could be the source of oxidation and degradation that may limit the durability of the models obtained.

The first layer is formed on an elevator platform, which is then lowered to allow new liquid resin to flow onto the working surface. This process is repeated, with each new layer adhering to the previous one. Once the model is finished, it undergoes a series of post-processes that makes it safe to handle. The insufficient extent of cure obtained during the building process means that it is necessary to complete the cure in an ultraviolet (UV) chamber during the so-called post-cure operation. Important consequences of insufficient polymerisation are warping due to relaxation, diffusion and evaporation of low-molecular weight components, and post-cure shrinkage due to density changes associated with the post-cure of the liquid resin trapped in the lattice structure (Flach and Chartoff, 1994; Weidemann *et al.*, 1995).

An alternative process to SL is proposed in this research paper. This process, designated by stereo-thermal-lithography (STLG), comprises both heat and UV radiation effects, solving some of the major limitations of conventional SL process, such as efficiency, speed, accuracy and tunability (Bartolo, 2001). The fabrication conditions used in this new process were established from the understanding of the physical and chemical transformations of thermosetting resins under thermal and photo-initiated curing reactions.

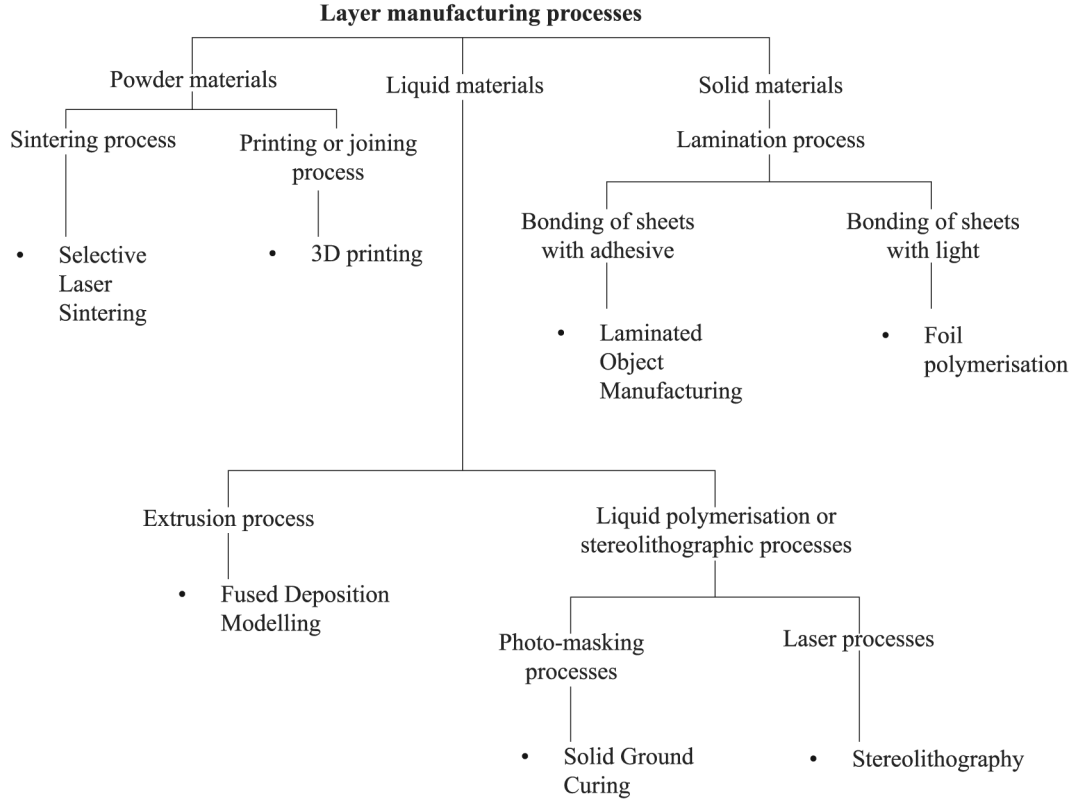
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Figure 1 Classification of rapid prototyping technologies (Campbell and Martorelli, 2000; Kruth, 1991)



The solidification process

The curing reaction of a photosensitive polymer is a polymerisation process characterised by chemical cross-linking reactions that create an infusible, insoluble and highly cross-linked three-dimensional network (Bartolo, 2001). During this reaction, two main events occur (Lange *et al.*, 2000; Prime, 1997):

- gelation, i.e. liquid-to-rubber transition, and
- vitrification, i.e. liquid or rubber-to-glass transition.

Gelation is a non-reversible event and corresponds to the incipient formation of an infinite molecular network, which is associated with a dramatic increase of viscosity (Prime, 1997). At this point, two phases coexist: a gel phase and a sol phase. The gel phase is the gelled part, insoluble in non-degrading solvents while the sol phase, which remains soluble, can be extracted with solvents. From gelation, as the reaction further progresses, the amount of sol phase presented in the system decreases and the polymer becomes more and more cross-linked.

Vitrification corresponds to the formation of a glassy solid material due to an increase in both the cross-linking density and the molecular weight of the polymer being cured, and usually follows gelation (Lange *et al.*, 2000; Prime, 1997). From vitrification, the rate of reaction will undergo a significant decrease and the reaction becomes very slow as it becomes controlled by the diffusion of the reactive species. The diffusion-controlled effect, besides causing a slow-down of the conversion reaction, determines the final degree of conversion obtained. Moreover, this effect may lead to the formation of non-homogeneous structures and alter the physical properties and durability of the cured polymer (Lovell and Bowman, 2003). It also creates changes in the specific volume of a material contributing to stress generation. Therefore, it is important to delay the vitrification effect, which avoids post-curing operations, in order to obtain better end products.

Light-molecules' interaction

Since most monomers or pre-polymers do not produce initiating species upon light exposure, it is necessary to introduce

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low-molecular weight organic molecules called photo-initiators that will start polymerisation, through photophysical and photochemical processes, generating reactive species such as radicals for free radical curing reactions or cationic species for cationic curing reaction (Fouassier, 1995). In this research, only free radical curing reaction mechanisms were investigated.

When a molecule absorbs light, electrons are set into motion by the oscillating electric field of the light. As the incident photon exchanges its energy with the molecule, a valence electron is promoted from the highest occupied molecular orbital to an unoccupied molecular orbital, with the formation of an excited singlet state molecule, which is an electronic state where the molecules possesses all paired electrons (Fouassier, 1995; Gilbert and Baggot, 1991). However, this excited molecule is a short living species that disappears by various competitive processes, dissipating the excited energy (Fouassier, 1995; Gilbert and Baggot, 1991). The absorption of light by a molecule and the subsequent evolution of its excited states are shown in Figure 2 through the Jablonski energy diagram. Two processes can be identified: photo-physical processes (radiative and non-radiative) and photochemical processes (Fouassier, 1995; Gilbert and Baggot, 1991).

Radiative mechanisms involve the absorption of a photon by a molecule in its ground state (S_0) and the emission of a photon from an electronically excited state by either fluorescence (de-excitation of an excited state with the same spin multiplicity of the ground state) or phosphorescence (de-excitation of an excited state with different spin multiplicity of the ground state). Non-radiative processes include internal conversion (IC) and the intersystem crossing (ISC), which occurs among the states of different spin multiplicity creating excited triplet states (T_1) that corresponds to electronic states where the molecule possesses two unpaired electrons (Gilbert and Baggot, 1991).

The photochemical process is a transformation of the starting molecule, through cleavage processes, electron transfer reactions, hydrogen abstraction, etc. Due to the short life of the excited singlet state molecules, many photochemical reactions occur only via excited triplet states, which are longer living species (Gilbert and Baggot, 1991; Guillet, 1985).

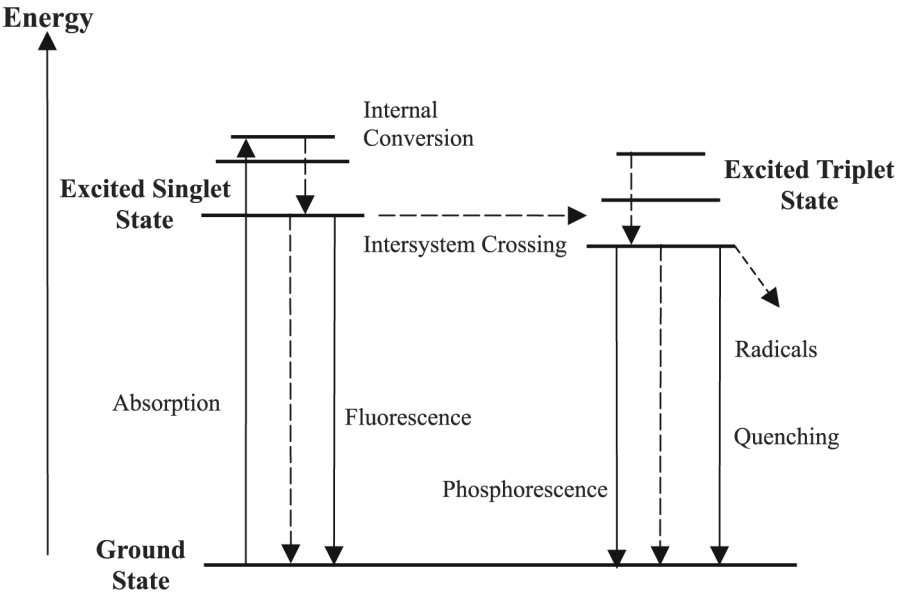
The SL chemical principle

In stereolithographic processes, resin samples containing a certain amount of photo-initiator are irradiated using UV radiation. The interaction of a photon of UV radiation with the photo-initiator can give one of the two results:

- the photon is not absorbed,
- the photon is absorbed.

In the latter case, the initiator molecule forms an excited singlet state, which then

Figure 2 The Jablonski energy diagram



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undergoes intersystem crossing to form a triplet state. This triplet state will generate radicals by undergoing α-cleavage fragmentation (Fouassier, 1995). The excitation and fragmentation process can be described by:

$$S_0 \xrightarrow{h\nu_1} S_1 \xrightarrow{ISC} T_1 \xrightarrow{\alpha-\text{cleavage}} \text{Radicals} \quad (1)$$

Finally, these radicals can undergo three different possible reactions:

- the radicals recombine to reform the photo-initiator molecule (reaction 1),
- the radicals escape from each other but are quenched by the inhibitor or oxygen molecules (reaction 2),
- the radicals are not quenched and react with monomers to start the curing reaction (reaction 3).

Among these three possible mechanisms, reaction 2 is the dominant one during the first stage of the curing reaction (Selli and Bellobono, 1993). This period is called the induction time and can be reduced by increasing the light intensity, photo-initiator concentration or temperature.

Optimisation of this subsystem can be achieved by balancing the effects of photo-initiator concentration and light intensity. However, it is important to point out the following.

- The increase in photo-initiator concentration increases the initiation and reaction rates and consequently the amount of solid material formed, reducing the diffusion controlled effects over the curing reaction delaying vitrification and avoiding post-cure operations. However, the increase in photo-indicator concentration reduces the UV penetration depth, according to the Beer-Lambert law, which means thinner layers and higher recording time.
- The increase in light intensity increases the initiation and reaction rates, increasing both the UV penetration depth and the speed of the process. Nevertheless, systems using high power UV laser beams are more expensive (Chen *et al.*, 1992).
- If the balance between photo-initiator concentration and light intensity is not the ideal one, this could also mean slow reaction rates. In this case, diffusion may be an additional problem (the reaction becomes less localised), with implications in terms of the accuracy of the models obtained.

The STLG chemical principle

The STLG process, shown in Figure 3, uses UV radiation and thermal energy (produced, for example, by infrared radiation) to initiate the polymerisation reaction in a medium containing both photo- and thermal-initiators.

The concentrations of both initiators are carefully selected and the reaction starts only when there is a combination of UV radiation and thermal energy. This way, the amount of each initiator must be low to start the polymerisation by only one of these two effects. However, at the point where the two effects intersect each other, the amount of radicals generated is sufficiently high to initiate the polymerisation process. On the other hand, the temperature is used to produce radicals through the fragmentation of thermal-initiators and simultaneously to increase the initiation and reaction rate of the photo-initiated curing reaction. Consequently, the extent of cure is increased, so no post-cure is needed. This is an important advantage of this process, because if products of high quality are required it is usually necessary to ensure a fairly high degree of gel formation. Moreover, if a small amount of solid material is formed, a large number of unreacted molecules are still present, so it is possible that an excessive heat generation will occur during the post-cure phase, leading to the distortion and warping of the built model.

The initiation process in this new approach can be described as follows.

- Path 1 (thermal energy effects):

$$\text{Initiator} \xrightarrow{\text{heat}} \text{Radicals} \rightarrow \text{Radical recombination and/or inhibition effects} \quad (2)$$

- Path 2 (UV radiation effects):

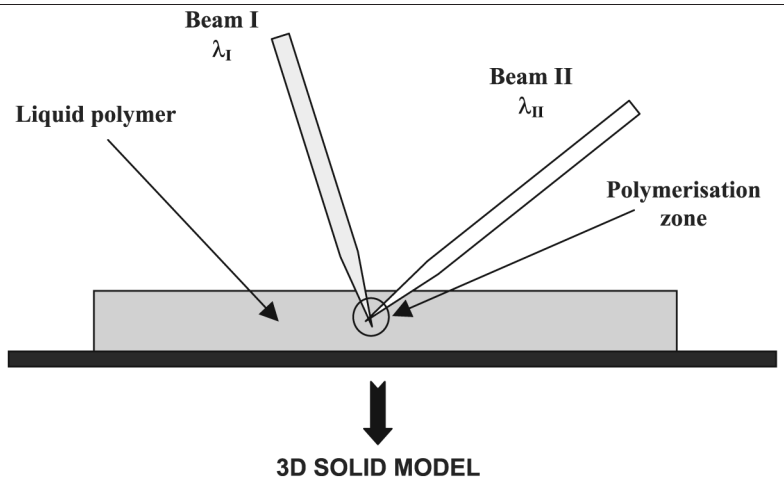
$$\text{Initiator} \xrightarrow{\text{UV radiation}} \text{Radicals} \rightarrow \text{Radical recombination and/or inhibition effects} \quad (3)$$

- Intersection of paths 1 and 2 (thermal energy + UV radiation effects):

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Figure 3 The dual laser curing principle of the STLG



$$\left.\begin{array}{l}\text{Initiator} \xrightarrow{\text{heat}} \text{Radicals} \\ \text{Initiator} \xrightarrow{\text{UV radiation}} \text{Radicals}\end{array}\right\} \rightarrow \begin{array}{l}\text{Sufficient number of radicals} \\ \text{to start the polymerisation}\end{array} \quad (4)$$

The chemical principle of this system corresponds to a different approach from the Swainson's previous one (Swainson, 1977), who tried to develop a stereolithographic system using radiation of two different wavelengths to write inside a vat containing a liquid resin. The work of Swainson presupposes that the curing reaction is initiated through the sequential bi-photonic excitation of a single type of initiator molecule. Different bi-photonic excitation processes were subsequently tested and the most important among them, in terms of the initiator molecules used, was the following one (Swainson, 1977; Swainson and Kramer, 1981, 1982, 1984):

$$S_0 \xrightarrow{h\nu_1} S_1 \xrightarrow{\text{ISC}} T_1 \xrightarrow{h\nu_2} T_n \quad (5)$$

In this case, the initiator molecule absorbs a photon of an appropriate wavelength and forms an excited singlet state, which then undergoes intersystem crossing to form the triplet state. This triplet state decays into the ground state if not further stimulated. However, if it is exposed to radiation of appropriate energy, it is capable of forming a higher excited triplet state, which is highly reactive.

The excitation mechanisms tested by Swainson presented two important problems. If a high concentration of initiator was used, the penetration depth of the radiation reduced and he was not able to write inside the polymeric medium. However, if low concentrations of initiator were employed, the penetration depth of the radiation increased, but the reaction became slow and less localised. Swainson was never able to solve these problems.

This new system presents several advantages with respect to conventional SL and the Swainson bi-photonic process (Bartolo, 2001):

- the generation of radicals is more efficient,
- small concentrations of the two types of initiator are used, enabling the radiation to penetrate deeper into the polymer,
- the combination of UV radiation and temperature increases the overall reaction rate avoiding post-cure operations,
- the curing reaction is more localised, which will improve the accuracy of the produced models,
- the system has more tunability.

This new fabrication approach represents a better system in terms of both the speed of the process and the quality of the models produced. It represents a truly one-step process for the generation of physical models because it avoids the time-consuming step of post-curing (performed in a separate apparatus). In addition, it represents a less expensive system compared with conventional SL, as the entire fabrication process is carried out in a single apparatus.

Implementation and results

To test the STLG principles, an in-house device was made by machining an aluminium

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block. In the initial stage of this research, the mask-based method of irradiation was the selected method because it is easier to implement experimentally. The device was placed on a metallic structure, which supports a mercury lamp (Blak-Ray B100 AP, UVP Inc.), transmitting light in the range 300-400 nm with λ_{max} at 365.5 nm, and is equipped with thermocouples connected to a temperature controller. Temperature gradients are produced by using heating elements and a refrigeration system, which uses cold-water flow. Moreover, the light intensity can be controlled by changing the distance between the plate containing the apparatus and the upper level of the metallic structure where the UV lamp is placed.

Different shapes of polymer structures were produced using two different types of masks created on black card (Figure 4 and Table I), and placed 2 mm above the sample position.

The accuracy of the STLG process was evaluated by comparing the dimensions of the produced models with the dimensions of the corresponding mask as indicated in Table II. As no mechanism was used to define a regular thickness value, the accuracy is only investigated in terms of the lateral dimensions of the models, which should be as close as possible to the dimensions of the masks.

In this study, a liquid resin system containing 0.5 wt per cent of photo-initiator was irradiated at room temperature using the conventional UV irradiation mechanism, while the same liquid resin system containing 0.01 wt per cent of photo-initiator and 0.01 wt per cent of thermal-initiator were thermal- and photo-cured at 110°C using the STLG principle.

Figure 4 Mask types

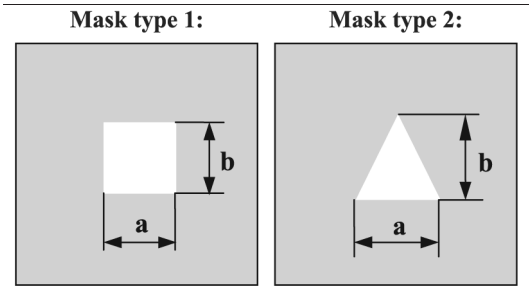


Table I Dimensions of each mask type

	Mask type 1	Mask type 2
Dimensions (mm)	$a = 11$	$a = 11.5$
	$b = 11$	$b = 11$

Table II Dimensions of models produced by conventional UV radiation and by STLG. The error is the difference between the object and the mask dimensions

	Dimensions (mm)	Error (mm)
Type 1		
Mask dimensions	11	
Object dimensions:		
Conventional UV irradiation	11.56	0.56
Stereo-thermal-lithography	11.23	0.23
Type 2		
Mask dimensions	$a = 11.5$; $b = 11$	
Object dimensions:		
Conventional UV irradiation	$a = 12.16$	0.66
	$b = 11.52$	0.52
Stereo-thermal-lithography	$a = 11.71$	0.21
	$b = 11.13$	0.13

The resin selected was a general-purpose unsaturated polyester (UP) resin (CRYSTIC 272 supplied by Scott Badder) containing 24 mol per cent isophthalic acid, 22 mol per cent fumaric acid, 29 mol per cent diethylene glycol and 25 mol per cent propylene glycol as determined by ^{1}H-NMR. The acid value is 18 mg KOH/g, and the number-average molecular weight is 3,111 g/mol. The styrene content (≈ 37 wt per cent) was determined by evacuating the styrene monomer in the vacuum oven at room temperature. The detailed composition of the resin used in this study is given in Table III.

The photo-initiator used was Irgacure 651 (Ciba-Geigy), which is the trade name for 2,2-dimethoxy-2-phenyl acetophenone. This is a low melting white crystalline powder initiator with $\lambda_{max} \approx 340$ nm as determined by UV-visible spectroscopy. When exposed to UV radiation this initiator undergoes a Norrish-type I photo-scission process (Hageman, 1989) forming two radicals.

Table III Composition of the UP resin used in this study

Composition (mol per cent):	
Isophthalic acid	24
Fumaric acid	22
Propylene glycol	25
Diethylene glycol	29
Molar ratio of glycols to dicarboxylic acids	1.2
Styrene content (wt per cent)	37
Number-average molecular weight (g/mol)	3,111
Unit molecular weight/mol polyester C═C (g/mol)	435
Average number of C═C/polyester	7
Molar ratio of C═C bonds for styrene-polyester	2.5

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The thermal-initiator used was Lucidol CH50X (Akzo Nobel), which is a dibenzoyl peroxide, 50 per cent powder with dicyclohexyl phthalate. This initiator produces radicals by a homolytic dissociation process.

Conclusions

This paper describes new advanced principles for stereolithographic applications through a process called STLG. This new process involves two distinct chemical reactions occurring simultaneously, the thermal-initiated curing reaction initiated by heat and the photo-initiated curing reaction initiated by the UV irradiation. Moreover, the resin samples have small concentration of thermal- and photo-initiators, so they show a very low reaction rate when they occur separately. However, when the two effects intersect each other, the population of radicals produced will be sufficiently high to initiate the curing reaction with a higher reaction rate. On the other hand, the coupling of temperature and UV radiation induces an increase in temperature in the regions where both effects are present, with advantages in terms of reaction rate and fractional conversions, improving the process in terms of a better-localised curing reaction. Moreover, this process represents a truly one-step process for the generation of physical models avoiding the time consuming step of the post-curing performed in a separate apparatus.

The use of polymeric systems containing small amounts of initiators, enables light to penetrate deeper inside the polymeric medium reducing the recoating process, which is a time-consuming step in conventional SL. The principles used in this STLG process can be applied to a new commercial system to generate models inside a liquid medium, turning the building process into a real three-dimensional process.

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